CLASS 8 – CBSE PHYSICS QUESTION PAPER

Time Allowed: 1 Hour

Maximum Marks: 30

SECTION A: SHORT ANSWER QUESTIONS

(4 questions × 2 marks = 8 Marks)

Q1. Force & Pressure:

(a) Define **atmospheric pressure** and state its SI unit.

(b) Explain why sharp cutting edge tools exert significantly higher pressure than blunt ones, referencing the formula $P = \frac{F}{A}$.

Q2. Sound Wave Terminology:

Define the following technical parameters of a acoustic wave:

(a) **Amplitude** and its relationship with the loudness of sound.

(b) **Frequency** and its SI unit, explaining how it determines the **pitch (shrillness)**.

Q3. Friction Types:

Arrange **static friction, sliding friction, and rolling friction** in increasing order of their magnitude. Define **interlocking of irregularities** and explain how lubricants reduce friction.

Q4. Chemical Effects of Electric Current:

Define **electroplating** and **electrodes**. Name the electrolyte and the cathode material used when coating an iron object with copper.

SECTION B: MEDIUM ANSWER QUESTIONS

(4 questions × 3 marks = 12 Marks)

Q5. Optics & Reflection:

(a) State the two fundamental **Laws of Reflection**. *(1.5 marks)*

(b) If a ray of light strikes a plane mirror making an angle of 35° with the **glancing surface (mirror surface)**, calculate the angle of incidence (θ_i) and angle of reflection (θ_r). *(1.5 marks)*

Q6. Electrostatics & Natural Phenomena:

(a) Differentiate between a **conductor** and an **insulator** based on free electron mobility. Give one example of each. *(1.5 marks)*

(b) Explain the working mechanism of an **electroscope** in detecting electrostatic charges via the process of **charge transfer and repulsion**. *(1.5 marks)*

Q7. Acoustics & Noise Pollution:

(a) Differentiate between **audible (sonic)** and **infrasonic/ultrasonic** frequency bands for human hearing. State the exact frequency spectrum for human audibility. *(2 marks)*

(b) Explain how **noise pollution** can lead to physiological conditions such as **hypertension** and **tinnitus**. *(1 mark)*

Q8. Fluid Mechanics & Atmospheric Physics:

(a) What is **hydrostatic pressure**? State two factors on which the liquid pressure exerted at a point inside a fluid depends. *(2 marks)*

(b) Why do deep-sea divers wear specially reinforced high-pressure suits when diving to great oceanic depths? *(1 mark)*

SECTION C: LONG ANSWER & ANALYTICAL PROBLEMS

(2 questions $\times$ 5 marks = 10 Marks)

Q9. Wave Motion & Sound Calculations:

(a) A sound wave generated by an underwater sonar transceiver travels at a speed of $v = 1500$ m/s in water. It bounces off the seabed and returns to the detector after a time interval of $t = 2.4$ seconds. Calculate the exact depth (d) of the seabed using the echo formula $d = \frac{v \times t}{2}$. *(2 marks)*

(b) A vibrating string completes 1200 oscillations in 15 seconds. Calculate its:

1. **Frequency (f)** in Hertz (Hz) using $f = \frac{\text{Number of Oscillations}}{\text{Time}}$. *(1.5 marks)*
2. **Time Period (T)** in seconds (s) using $T = \frac{1}{f}$. *(1.5 marks)*

Q10. Human Eye Anatomy & Vision Defects:

(a) Define the role of the following anatomical components of the human eye:

1. **Cornea & Lens** (Refraction & Focusing)
2. **Iris & Pupil** (Aperture Control)
3. **Retina & Optic Nerve** (Photoreception via Rods/Cones & Signal Transmission) *(3 marks)*

(b) Explain the difference between **Myopia (nearsightedness)** and **Hypermetropia (farsightedness)** in terms of image focal point placement relative to the retina. Mention the type of corrective lens used for each defect. *(2 marks)*